Ex: 07
Date: 6/08/26

Flow Control at Data link layer using sliding window.

AIM :-

To write a program to implement flow control at Data link layer using sliding window protocol and to simulate the flow of frames from one node to another.

Algorithm :

1.) Start the Program
2.) Read the total number of frames to be sent (N), the sender window Size (w) and a frame number to simulate as lost during transmission.
3.) Initialize base = 0 (sequence no. of the oldest un-acknowleged frame) and next seq = 0 (sequence number of the next frame to be sent).
4.) Repeat while base < N :
   a.) Determine the current Sender window as the range [base, base + w-1]
   b.) Send all frames with sequence numbers from next seq to (base + w-1), increment next seq after each time / frame, since the window restricts the sender to atmost - w outstanding (unacknowledged) frame at any time.
   c.) If the lost frame lies within the frames just sent:
   (i.) The receiver discards the lost frame and any subsequent out of order frames.
   (ii.) The sender times out and retransmits all frames starting from lost frames.
5.) Repeat step 4 until all N frames have been transmitted.
6.) Stop the program.

SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATIO

Enter total number of frames to send : 8
Enter sender's window size : 2
Enter frame number to simulate as LOST
(0 to 7, enter -1 for no loss)
: 5

Sender window : [Frame 0 - Frame 1]
Node A ---> Sending Frame 0
Node A ---> Sending Frame 1
Node B ---> Received Frame 0, Sending ACK 0
Node B ---> Received Frame 1, Sending ACK 1
Node A ---> ACK 1 received. Sliding window forwa

Sender Window : [Frame 2 - Frame 3]
Node A ---> Sending Frame 2
Node A ---> Sending Frame 3
Node B ---> Received Frame 2, Sending ACK 2
Node B ---> Received Frame 3, Sending ACK 3
Node A ---> ACK 3 received. Sliding window forwa

Sender window : [Frame 4 - Frame 5]
Node A ---> Sending Frame 4
Node A ---> Sending Frame 5
Node B ---> Frame 5 missing. Discarding out of orde
frames after it.
Node A ---> Time out! Resending window starting from
window Frame 5.

Program :-

```
def Simulate (total_frames, window_size, lost_frame)
    base = 0
    next_seq = 0
while base < total_frames :
    window_end = min ( base + window_size - 1, total_frame
                                                -1).

    print ("\n----------")
    print ("f sender Window : [Frame {base} - frame {
                                    window_end} ]")
    Send all frames within current window that haven't
    been sent
    while next_seq < base + window_size and next_seq
                                    < total_frames :
    print ("f Node A --> sending frame {next seq}")
    if next_seq == lost frames :
    print (f"(Frame { next_seq } lost in (transit!)")
      next_seq += 1
    if lost_frame is not none and base <= lost frame
                                    < next seq
    print ("f Node B ---> Frame { lost_frame} missing)
    print ("f Node A ---> Time out ! Resend Frame)
    next_seq = lost frame
    Continue => lost frame = None
  # no loss in this batch
for f in range ( base, next seq):
    print ("f Node B ---> Received Frame {f}, sending)
    print (f" Node A ---> ACK {next_seq} received .
                    sliding window forward")
base = next_seq
print ("\n----------")
print (f" All { total_frames } frames successfully from
                                    A to d.
```

Sender window : [Frame 4 - Frame 5]

Node A ---> Sending Frame 5

Node B ---> Received Frame 4, sending ACK 4

Node B ---> Received Frame 5, sending ACK 5

Node A ---> ACK 5 received. Sliding window forward

Sender window : [Frame 6 - Frame 7]

Node A ---> Sending Frame 6

Node B ---> Sending Frame 7

Node B ---> Received Frame 6, sending ACK 6

Node B ---> Received Frame 7, sending ACK 7

Node A ---> ACK 7 received. Sliding window forward.

All 8 frames successfully transmitted frome Node A to Node B.

```
def main():-
    total_frames = int(input())
    window_size = int(input())
    lost_frame = int(input())
    if lost_frame == -1:
        lost_frame = None
    simulated total_frames, window_size, lost_frame
if __name__ == "__main__":-
    main()
```

Result :-

The program to implement flow control at the data link layer using the sliding window Protocol was successfully written in python, executed and verified by simulating the flow of frames between two nodes (Node A as sender and Node B as receiver).